Hyung Cho

Software Test Engineer | Embedded Product Validation, Python Test Automation, and Root-Cause Analysis

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SUMMARY

Validation engineer with 10 years at Zebra Technologies testing embedded hardware products: test plans, procedures, acceptance criteria, and final reports for 6 shipped products, and root-cause isolation across hardware, firmware, and software. Builds and maintains automated tests in Python, Swift, and C#, with quality gates in GitHub Actions CI. Directs AI coding agents that write code, tests, and failure diagnoses, with a written risk assessment on each change and CI checks that keep requirements traceable to test evidence.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Wrote an offline Python test suite of 314 test functions that runs on every pull request with credentials stripped and network access blocked, plus integration tests against a local Supabase stack; 39 of 39 modules pass in CI.
- Traced a failing LLM evaluation gate to the test's own judge rather than the product, then compared five judge designs on replayed outputs and cut wrong verdicts from 38 of 280 to 0 of 300.

Rep Today, native iOS workout app, pre-release

- Grew the XCTest unit suite from 829 to 1,700 tests; CI runs 1,678 of them on both iOS 18 and iOS 26 with 0 failures.
- Found and fixed platform-specific defects through CI and cross-OS testing, including an iOS 18 layout feedback loop that hung the suite past the 45-minute job limit, closed with a regression test.
- Directed development as 214 agent-authored pull requests, 195 with a written risk assessment; later ones separate scenarios validated live in the app from those covered only by tests.

Immerse, film-camera iPhone app, in development

- Added CI scripts that fail the build when the task tracker and the requirement evidence map disagree (154 requirements, 74 acceptance clauses, 9 invariants) or when a debug-only unlock reaches the release build.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Led a DeepLabV3+ cancer-cell segmentation pipeline: preprocessing, training, evaluation, and deployment.
- Fine-tuned LLaMA and Falcon, integrated ChatGPT into a chatbot, and iterated on prompts with an evaluation workflow.
- Containerized models with Docker for reproducible environments across volunteer teams in several time zones.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Led validation testing for 6 shipped hardware products, writing test plans, procedures, acceptance criteria, and final reports.
- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved each fix.
- Built C# validation tooling that replaced manual test workflows and cut regression test cycles by about 30%.
- Owned SDK validation across Windows, macOS, and Linux: cross-platform compatibility, performance, and reliability.

SKILLS

Validation: Test planning, Hardware validation, Failure analysis, Regression testing, Requirements traceability

Test automation and CI: Python, JavaScript, C#, Swift, XCTest, GitHub Actions, CI/CD, Shell scripting, Flaky test diagnosis

AI-assisted testing: AI coding agents, Claude Code, Code review, LLM evaluation, LLM-as-judge, Technical writing

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo

FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses